



Third Person Shooter Controller - Cover Add-on

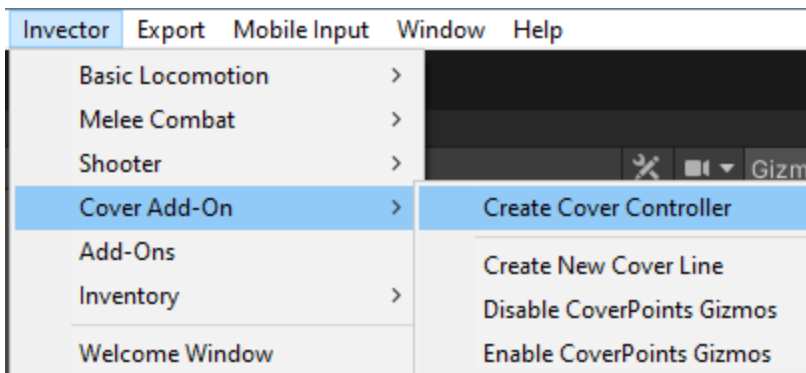
(version 0.7 beta)

Thank you for supporting this add-on, it's thanks to you guys that we continue to support our assets by creating new features, improvements, and bug fixes.

--- Invector Team ---

- CREATING A NEW COVER CONTROLLER

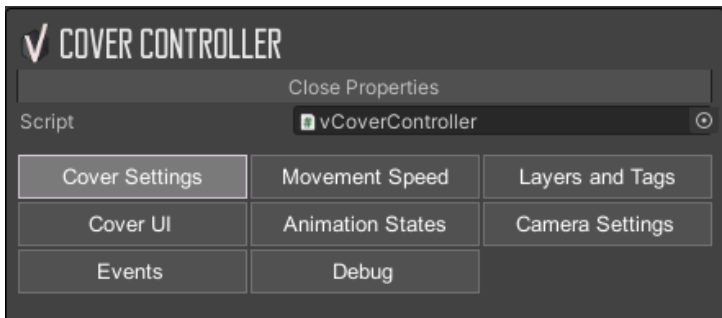
Open the Invector menu and go to *Cover Add-On > Create New Cover Controller*.



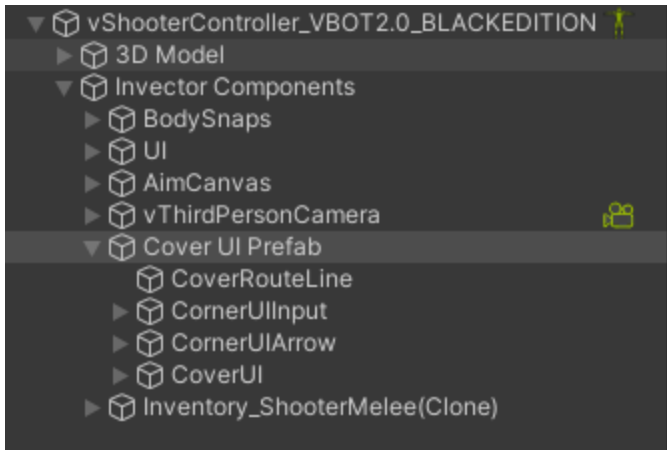
Assign your model and hit Create.



Notice that the Controller already comes with the **Invector@Cover Animator Controller** and the **Cover Controller** component assigned for you, it also contains the prefab **CoverUI** inside the player under the Invector Components gameobject.

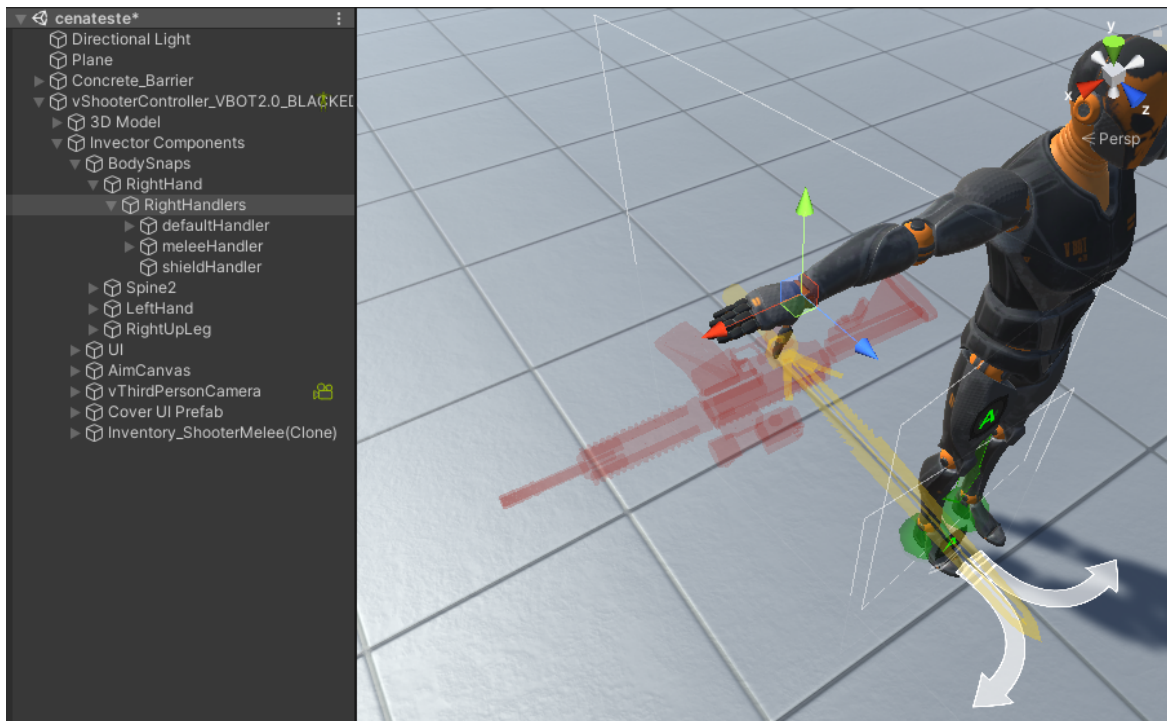


Here you can modify the Cover UI elements to your own design.



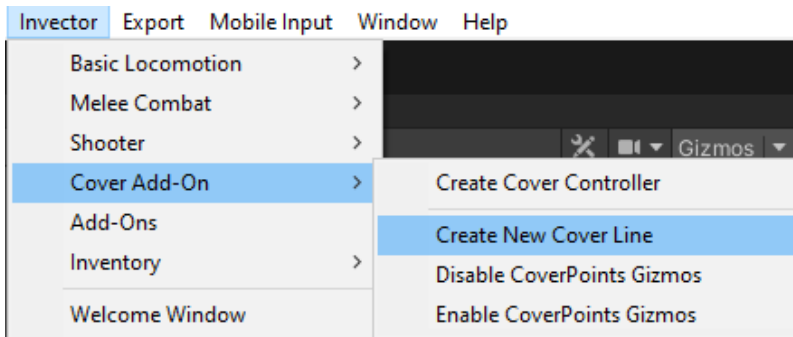
Align the Handlers on your character so that the Weapons are aligned correctly.

Video Tutorial on how to Align Weapons in depth: <https://youtu.be/cUgkkowMeSg?t=272>

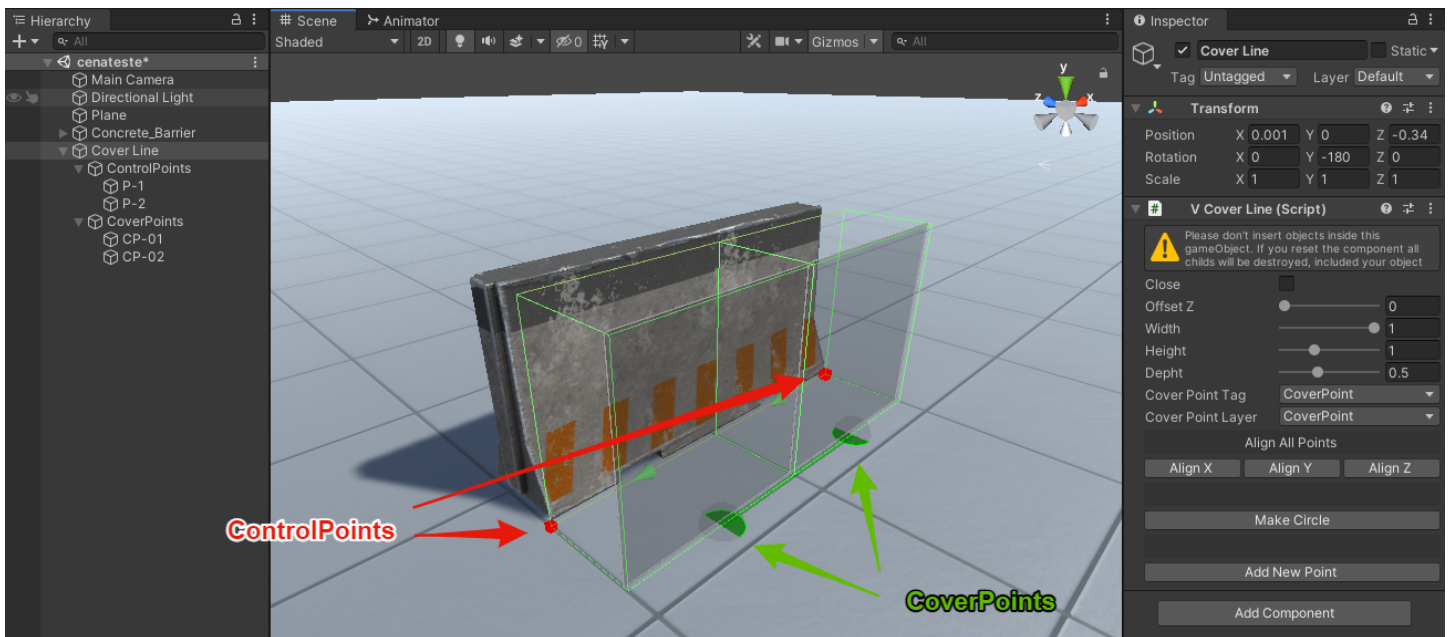


- CREATING A NEW COVER LINE

Open the Investor menu and go to *Cover Add-On > Create New Cover Line*.



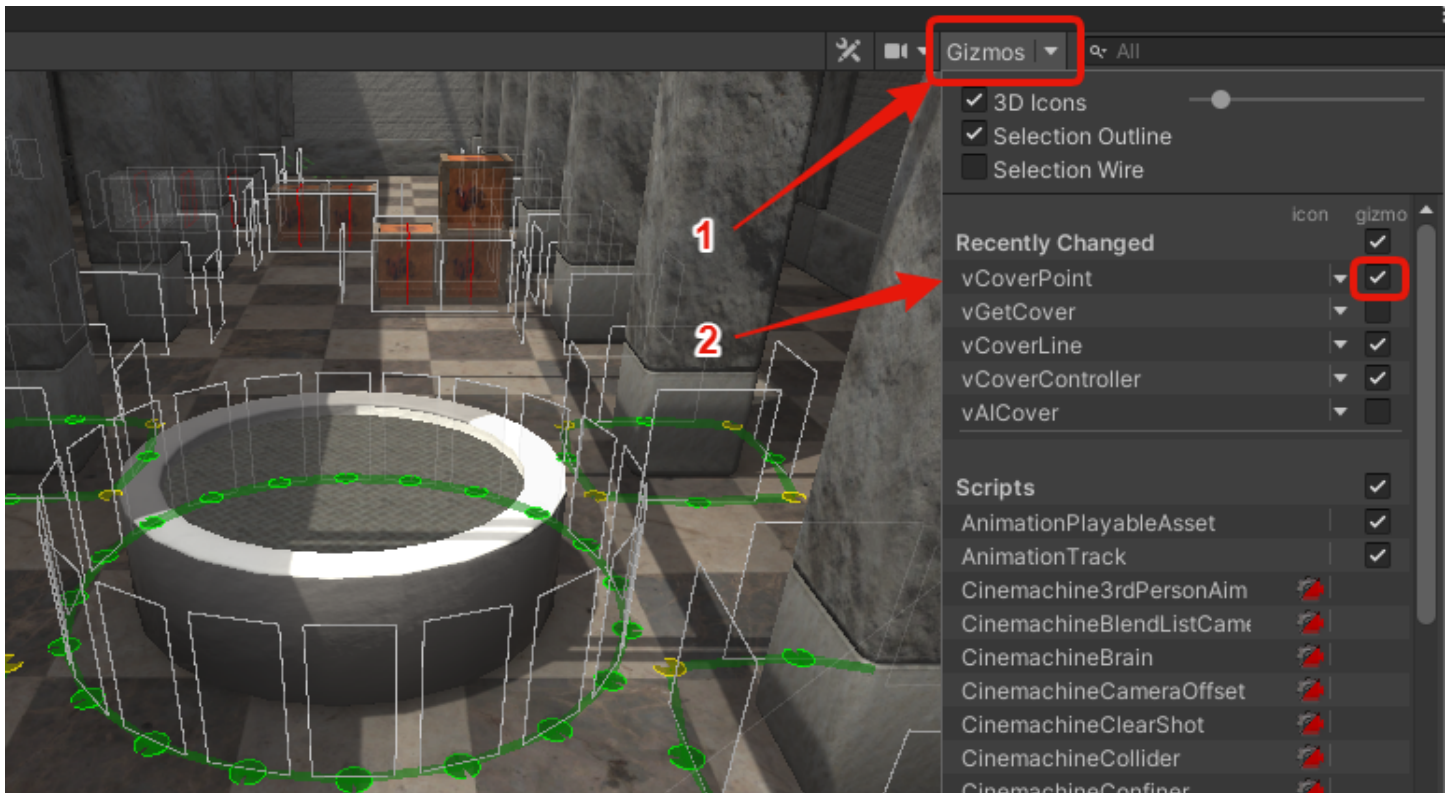
A CoverLine GameObject will be created with the vCoverLine Component, inside this object you will find other 2 called ControlPoints and CoverPoints.



- **ControlPoints** are used to control the length of your CoverLine, click on the red square gizmos and use the Transform tool to move it around to fit the area you want to cover.
- **CoverPoints** are used to identify cover locations available for the controller to go and enter cover mode. They also form the path that the player will walk on while in cover mode.
- **Close** is the option to automatically close a cycle of cover points
- **OffsetZ** is a offset of the Z axis
- **Width** controls the width of the collider, lowering will create more colliders between the space determined by the control points.
- **Height** will adjust the height of the colliders

- **Depth** controls the depth of the collider, make sure to always create a depth higher than the capsule collider radius of your Character Controller.

To see the ControlPoints and CoverPoints you need to **Enable the Gizmos [1]** in the SceneView



The Gizmos from the vCoverPoint are quite expensive on performance so make sure to **TURN OFF [2]** when you're done editing your CoverPoints, otherwise it will affect performance specially during PlayMode without Maximize the Game Window.

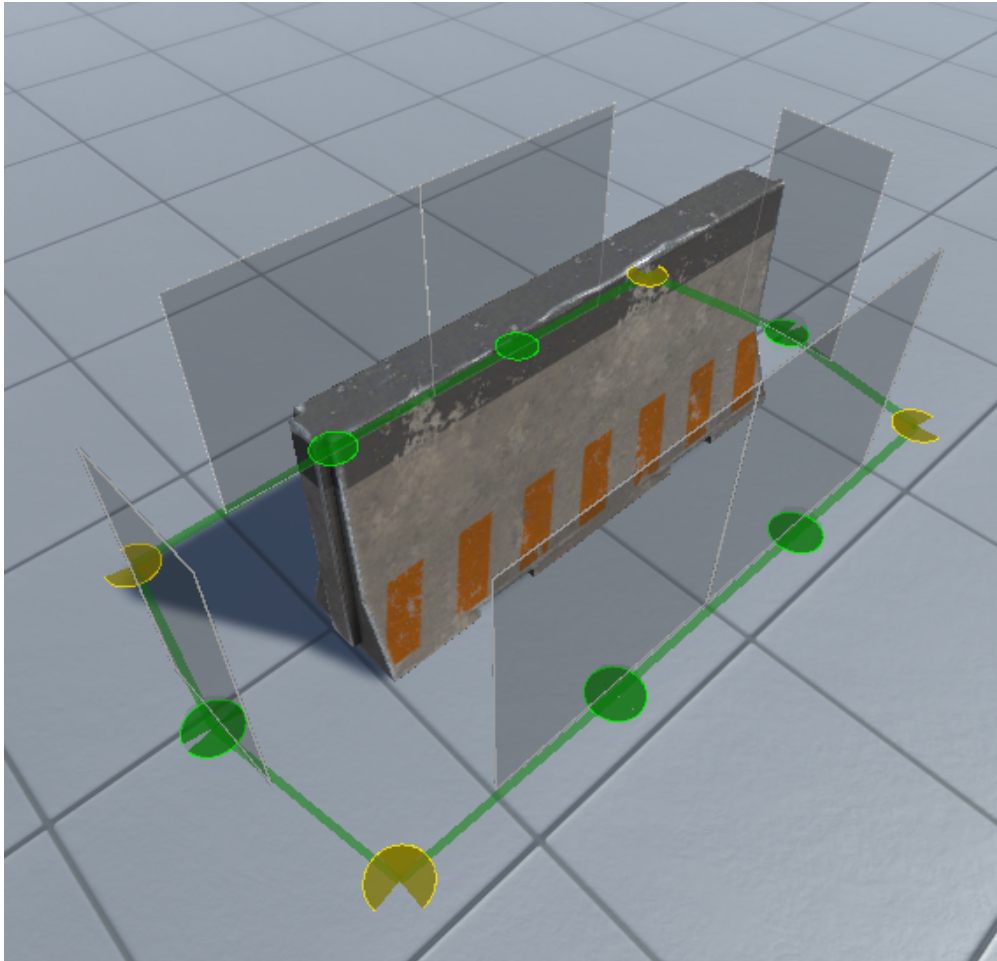
Cover Add-On	>	Create Cover Controller
Add-Ons		Create New Cover Line
Inventory	>	Disable CoverPoints Gizmos
Welcome Window		Enable CoverPoints Gizmos

You can also enable/disable the CoverPoints Gizmos in the Invector menu

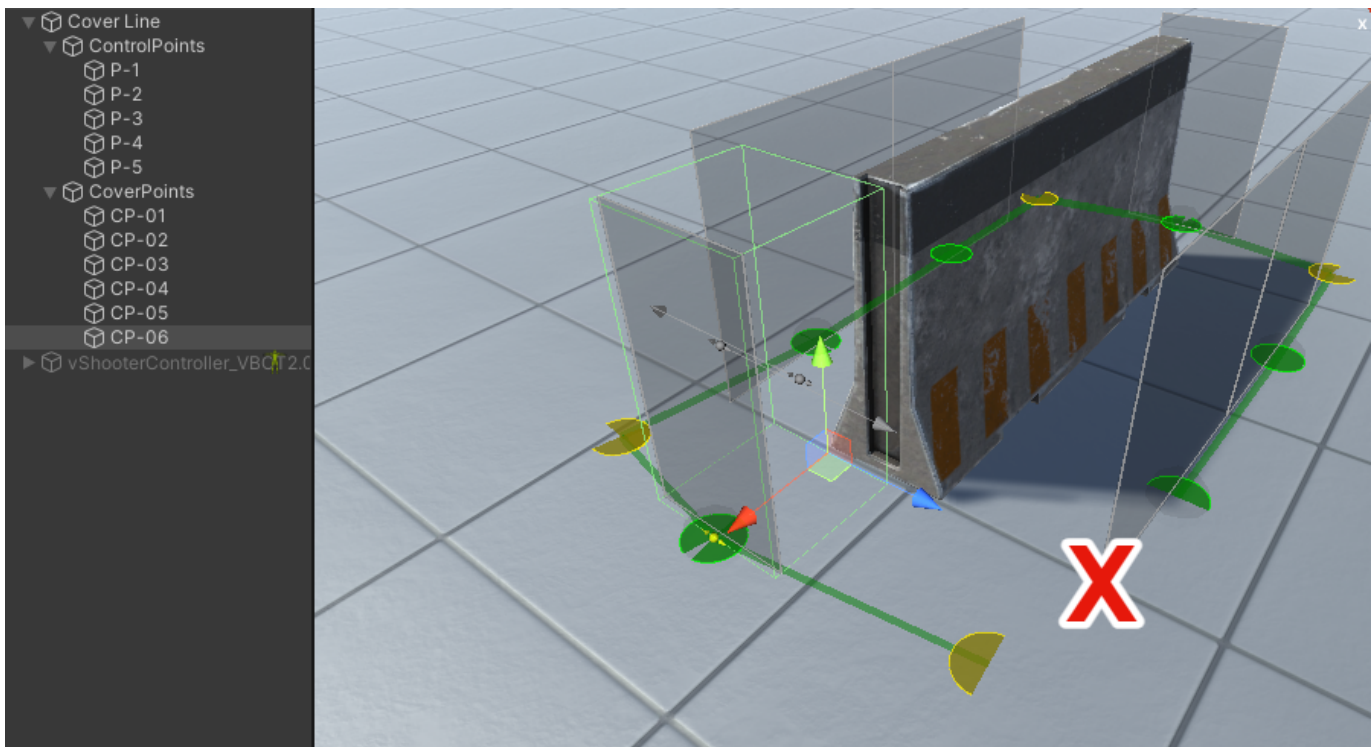
* IMPORTANT *

You must **Bake the Navmesh** of your scene to see the ControlPoints Line, the CoverController uses the Navmesh to move the player automatically when entering from a distance or moving between distant CoverLines, it's also a good idea to add [NavMeshObstacles](#) and enable the option *Carve* to your Cover Prefabs so you don't have to keep baking the scenes everytime you move the prefab.

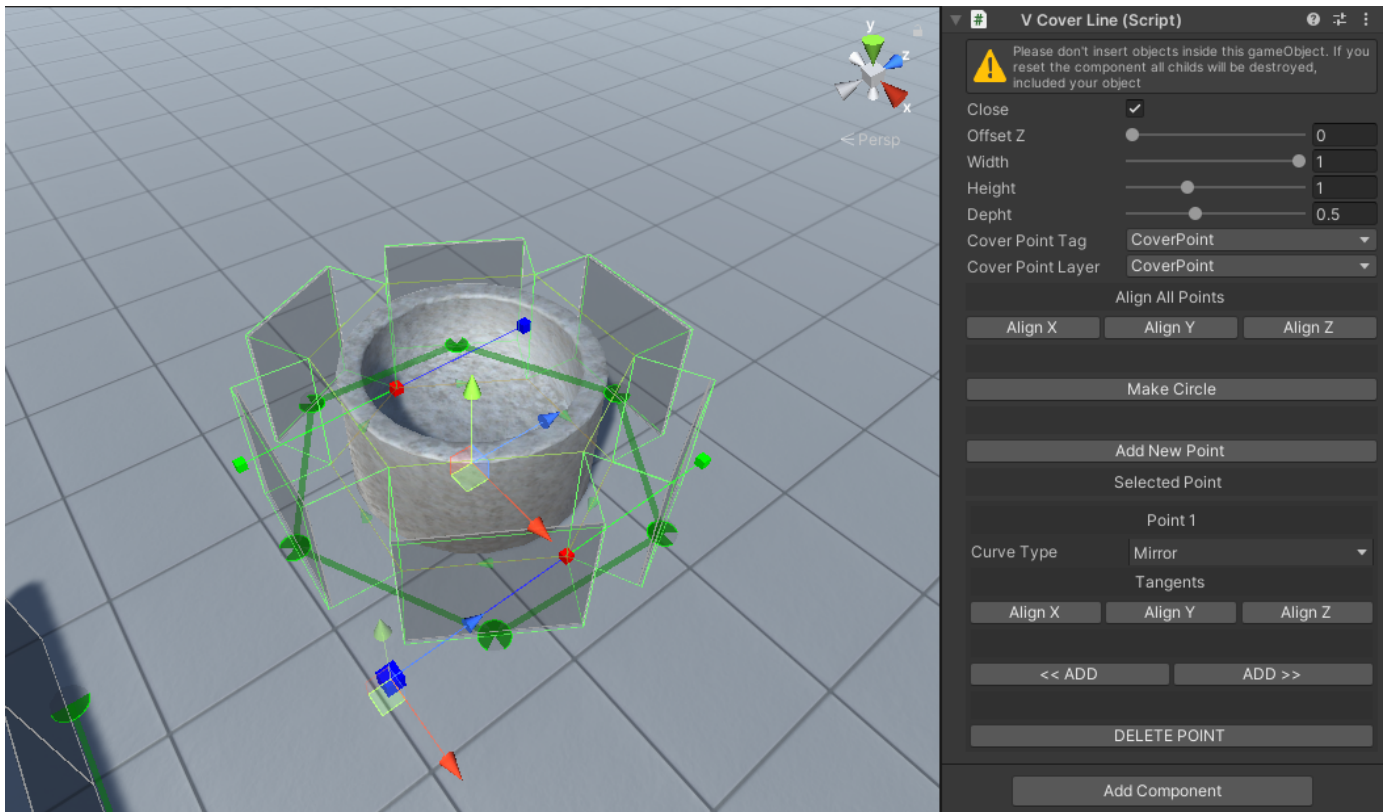
You can create new Points by clicking on the “*Add New Point*” button, this point is separated from the other group so you rotate individually to create corners for example.



You can also select and move a ControlPoint child to adjust better on your model, but make sure the line is not broken, otherwise the player won't be able to walk on it

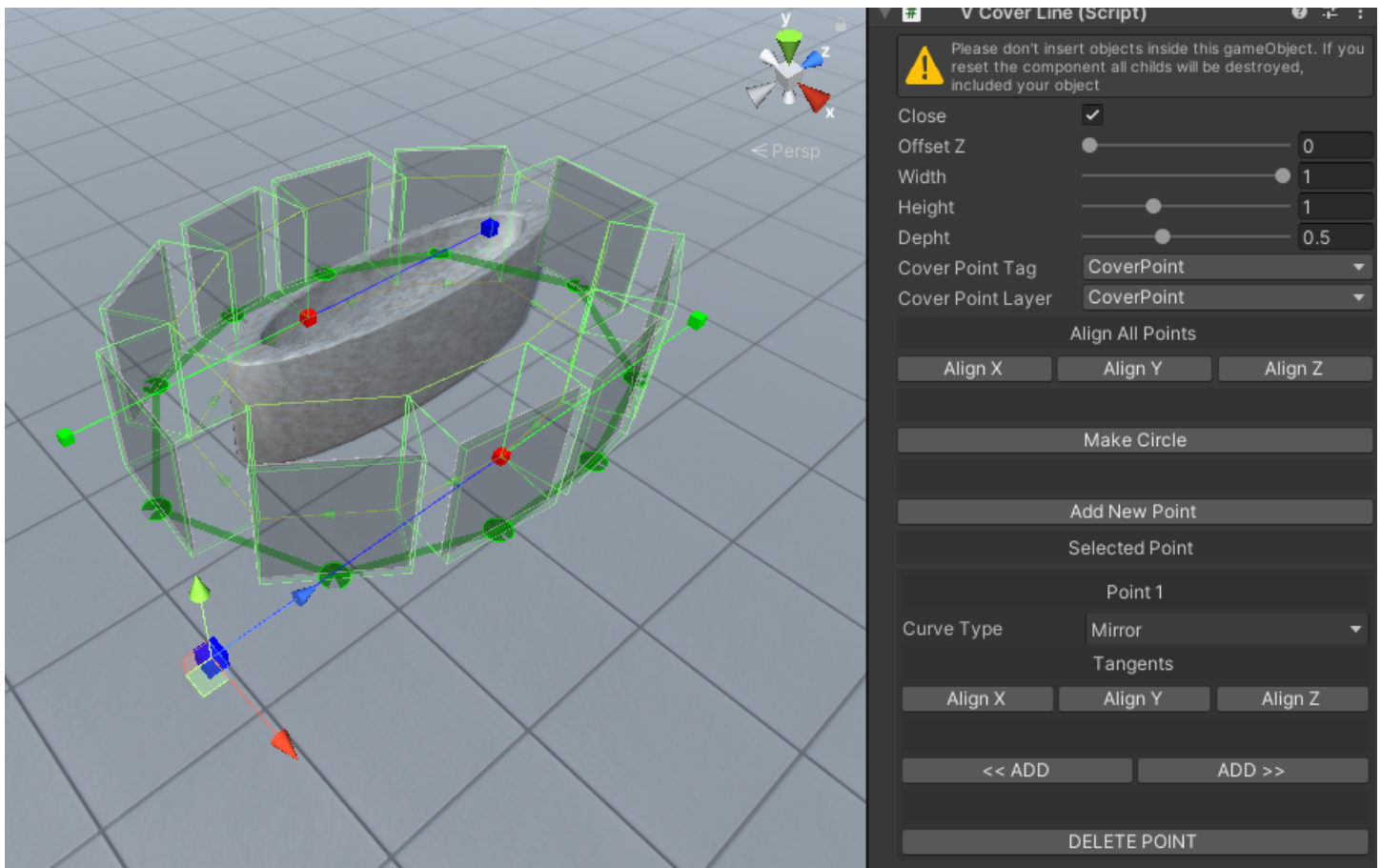


The CoverLine Component has some cool options such as **Make Circle** which will automatically create a Cover on a Circle formation for you



You can also **Align X, Y and Z** to All Points or Specific Points by using their respective Align buttons.

When you create a Circle some new blue and yellow points will appear in which you can control **Curves** to adjust better on your 3D Models, there are also several **CurveTypes** which you can choose from.



* IMPORTANT *

Again, don't forget to **Bake Your Navmesh** otherwise the CoverPoints won't work correctly, another tip is to add a [NavMeshObstacle](#) and enable the option **Carve** into your Cover Prefabs so you don't have to keep baking the scene everytime you move a cover prefab from its location.

- COVER SETTINGS

Cover Settings	Movement Speed	Layers and Tags
Cover UI	Animation States	Camera Settings
Events	Debug	

Enter/Exit Cover Input

USE	Enter Exit Input
Auto Enter Cover	☒
Enter Input Direction Angle	60
Enter Direction Timer	0
Auto Exit Cover	☒
Exit Input Direction Angle	160
Exit Direction Timer	0

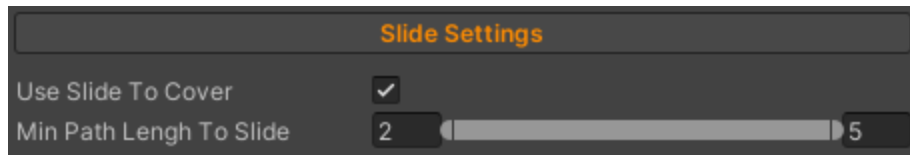
- **Enter/Exit Input:** Check to Use and assign an input to Enter and Exit the Cover, this input will be ignored if the option AutoEnterCover is enabled.
- **AutoEnterCover:** Check this option to automatically enter cover mode when close to a cover point without input and within the EnterInputDirectionAngle limit.
- **EnterInputDirectionAngle:** Angle of Input direction to enter cover
- **EnterDirectionTimer:** delay to enter cover
- **AutoExitCover:** Automatically exit the cover when pressing the input within the ExitInputDirectionAngle value.
- **ExitInputDirectionAngle:** Angle of Input direction to exit cover
- **ExitDirectionTimer:** delay to exit cover

Get Cover Settings

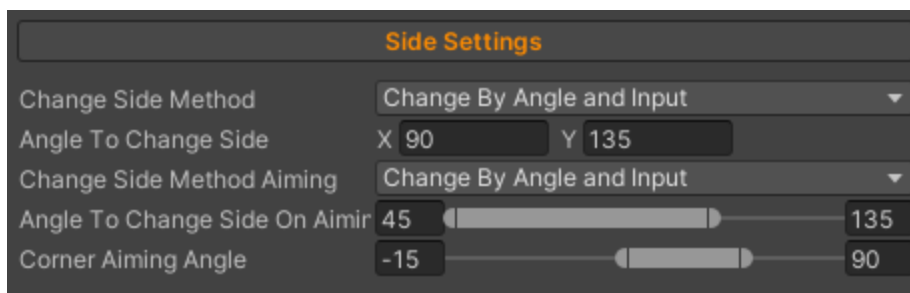
Max Enter Cover Distance	<input type="range" value="4"/>
Min Distance To Cancel Travel	2.5
Auto Travel To Next Cover	☒
Next Cover Distance	<input type="range" value="4"/> 10
Ray Cast Angle Pass	<input type="range" value="0"/>
Ray Cast Angle Multiplier	<input type="range" value="1"/>
Capsule Radius In Cover	0.2
Min Angle Of Valid Cover	<input type="range" value="80"/>
Position Offset Z	0.01

- **maxEnterCoverDistance:** Distance to get cover point to enter in cover - Only works with Auto Enter disabled
- **mindDistanceToCancelTravel:** The minimum distance to cancel moving to a CoverPoint if you release the input

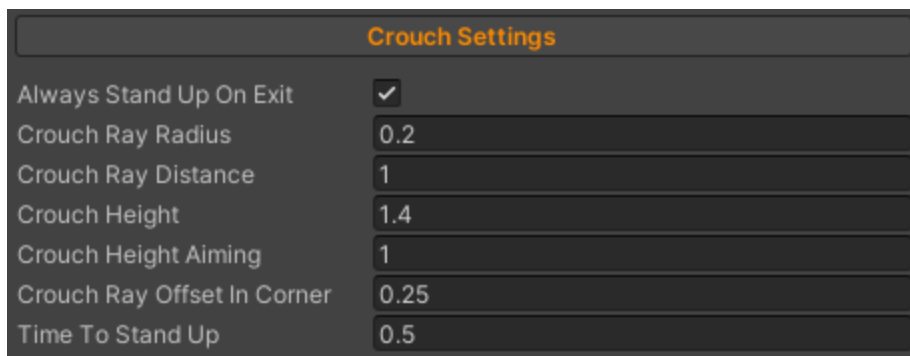
- **autoTravelToNextCover:** Enable to Detect distante CoverPoints and Travel to it *NAVMESH REQUIRED
- **nextCoverDistance:** Min Max Distance to get a new cover point when it's in cover. The Min Value is ignored if it is in cover and next Cover is not connected to the current cover line. Works with autoTravelToNextCover enabled
- **rayCastAnglePass:** Creates more Raycasts within a open angle, Turn on DEBUG to see it
- **rayCastAngleMultiplier:** Multiplies the raycastAnglePass into several casts, Turn on DEBUG to see it
- **capsuleRadiusInCover:** Capsule Collider Radius while in Cover
- **minAngleOfValidCorner:** The minimum angle accepted to be a valid corner
- **positionOffsetZ:** Distance OffsetZ of the Controller from the Cover



- **useSlideToCover:** Check to use a Slide animation when switching between covers using the AutoTravel
- **minPathLengthToSlide:** Min and Max distance to Slide



- **changeSide:** Choose how to change side, using input direction or angle and input
- **angleToChangeSide:** Angle X and Y to change side
- **changeSideAiming:** Choose how to change side while aiming, using input direction or angle and input
- **angleToChangeSideOnAiming:** Angle X and Y to change side when aiming
- **cornerAimingAngle:** Angle to keep in corner state pose when aiming



- **alwaysStandUpOnExit:** Standup when exiting a Cover Crouched or Keep it crouched
- **crouchRayRadius:** Radius to verify if the character will be on crouch, Turn on DEBUG to see it
- **crouchRayDistance:** Distance to verify if the character will be on crouch, Turn on DEBUG to see it

- **crouchHeight:** Height to verify if the character will be on crouch, Turn on DEBUG to see it
- **crouchHeightAiming:** Height to verify if the character will be on crouch aiming, Turn on DEBUG to see it
- **crouchRayOffsetInCorner:**
- **timeToStandUp:** delay to stand up

Corner Settings	
Enter Using Get Cover Input	☐
Angle To Enter Corner	-10
Min Angle To Start Move To Co	10
Time To Enter Corner	0.5
Corner Left Aiming_Offset Positi	0.6
Corner Right Aiming_Offset Posi	0.4
Corner Weight Smooth	20

- **enterUsingGetCoverInput:** Check to use the GetCoverInput to move on corners, Keep it False to just press the input direction to do corners, this will also change the Corner UI
- **angleToEnterCorner:** Angle to Enter a Corner
- **minAngleToStartMoveToCorner:** Minimum angle to start moving to a corner
- **timeToEnterCorner:** Delay to do Corner
- **cornerLeftAiming_OffsetPosition:** Controller Offset Position when Aiming on a Corner
- **cornerRightAiming_OffsetPosition:** Controller Offset Position when Aiming on a Corner
- **cornerWeightSmooth:** Smooth when performing a Corner

Generic Actions Settings			
Possible Actions			
Size	1		
Element 0	Action		
Check Action Margin	X 0.3	Y 0.2	Z 0.4
Check Action Offset	X 0	Y -0.1	Z -0.2
Show Action Gizmos	☐		

Here you can add Custom Actions into Covers, for example, if you have a barricade but also want to perform a vault (jump over) over it, just add the Action name to the list.

- COVER MOVEMENT SPEED

Cover Settings	Movement Speed	Layers and Tags
Cover UI	Animation States	Camera Settings
Events	Debug	

Movement Speed	
Crouched Move Speed	2
Standing Move Speed	2
Go To Cover Point Running Move Speed	4
Go To Cover Point Sprinting Move Speed	6
Go To Corner Crouched Move Speed	2
Go To Corner Standing Move Speed	2
Rotation Speed Moving	5
Rotation Speed Idle	10

Animation Speed	
Stand Animator Speed	1.2
Crouch Animator Speed	1.6
Go To Cover Point Running Animator Speed	1
Go To Cover Point Sprinting Animator Speed	1
Go To Corner Point Animator Speed	1.5

Here you can set custom movement speed for the controller and the animation itself separately.

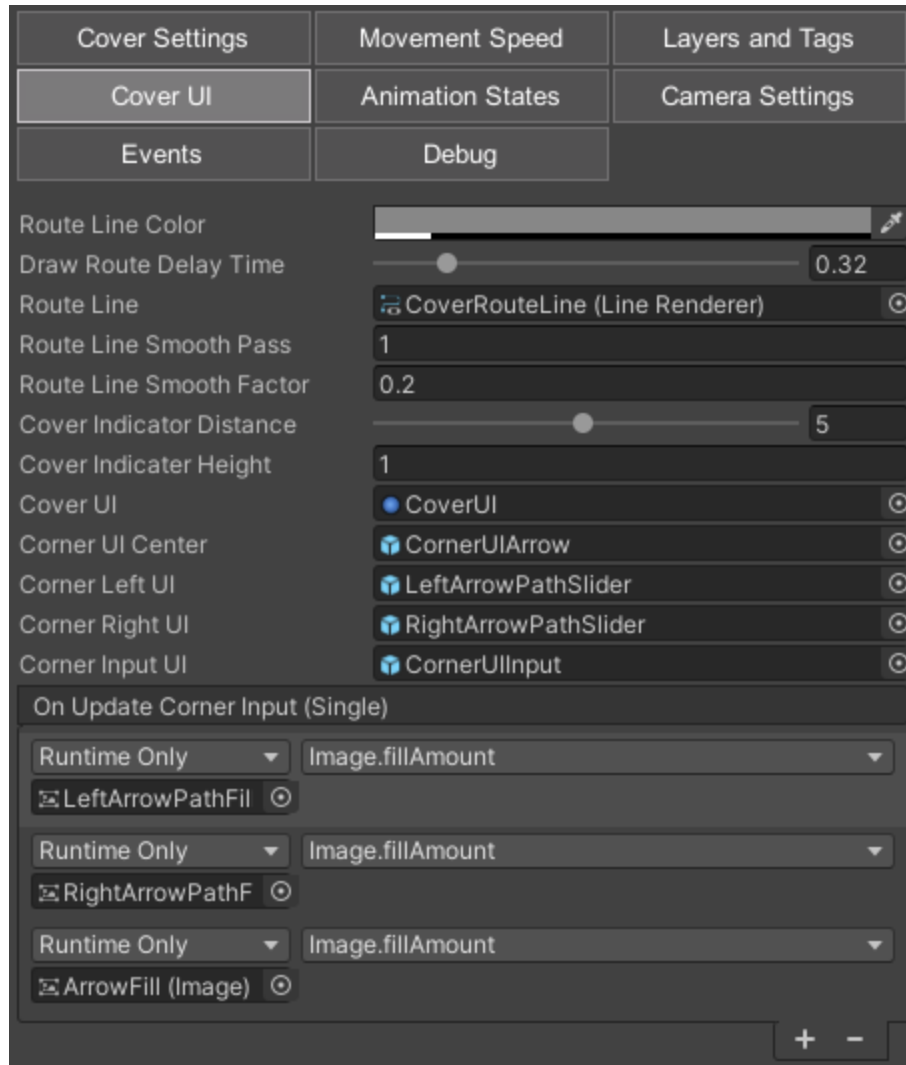
- COVER LAYER AND TAGS

Cover Settings	Movement Speed	Layers and Tags
Cover UI	Animation States	Camera Settings
Events	Debug	

Obstacles Layer	Mixed...
Check Height Ignore Tags	Glass
Check Height Ignore Names	
Size	1
Element 0	glassFrame
Cover Layer	CoverPoint
Cover Tag	CoverPoint

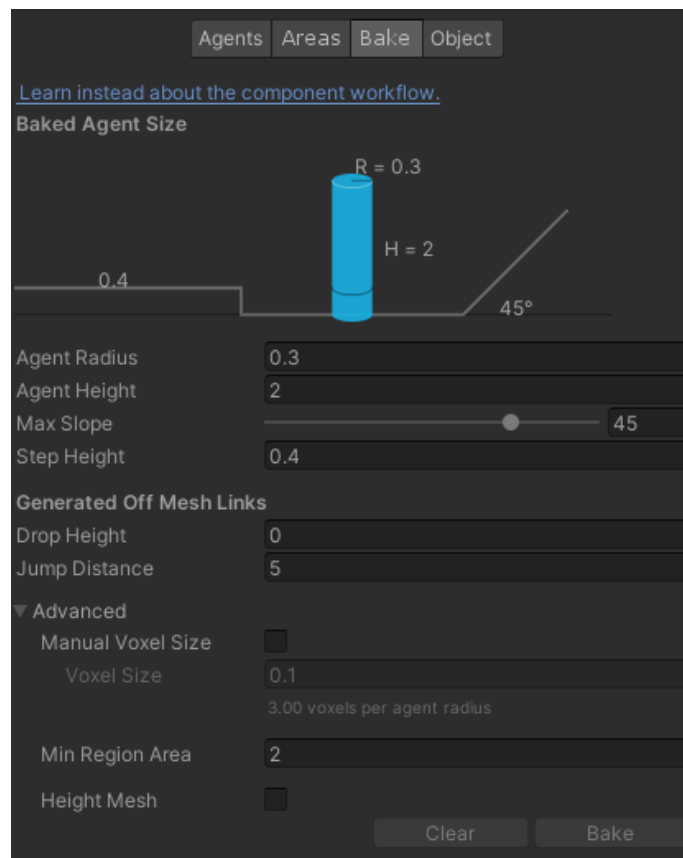
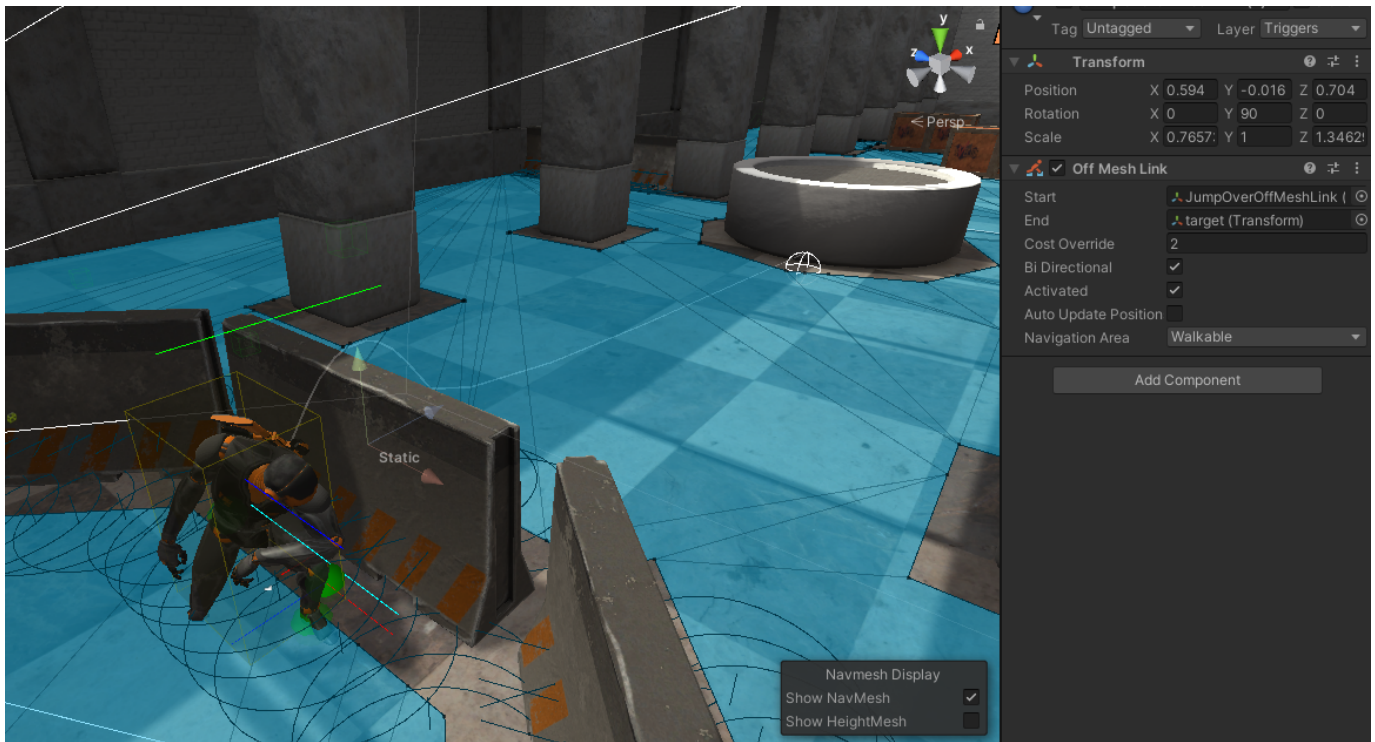
- **obstaclesLayer:** You can assign different Obstacle Layers for your different objects in the scene
- **checkHeightIgnoreTags:** Filter tags that will be ignored by the Crouch verifications and stay crouched.
- **checkHeightIgnoreNames:** Filter gameObject names that will be ignored by the Crouch verifications and stay crouched.
- **coverLayer:** Layer of the cover must be the same as the CoverPoint inside your CoverLine)
- **coverTag:** Tag of the cover must be the same as the CoverPoint inside your CoverLine)

- COVER UI



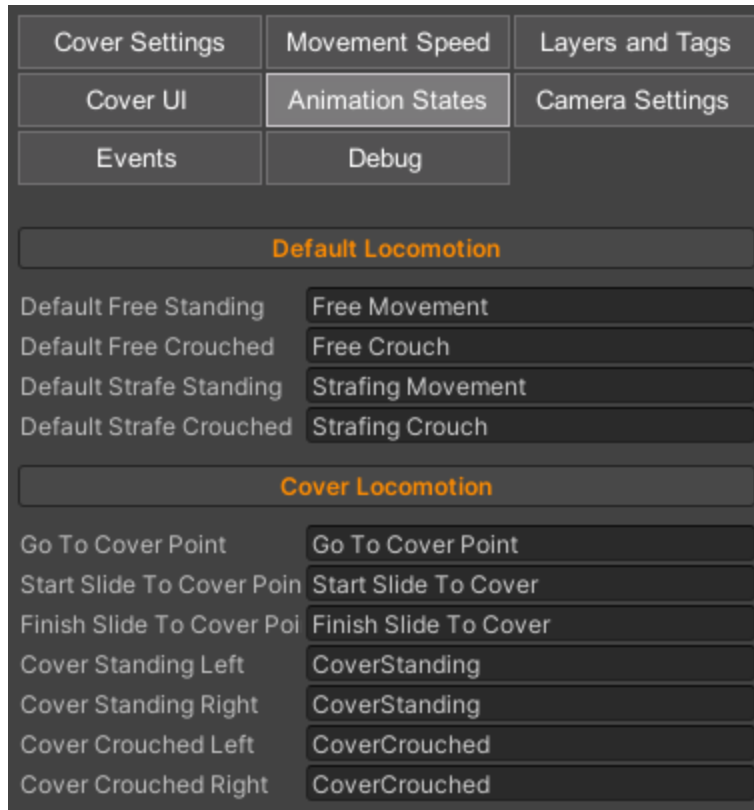
Here you can customize some settings about the Cover UI, including the Route Path, CoverInput and Corner. The Events and prefabs will be automatically assigned once you create the Cover Controller using the Cover Controller Creator Window.

To create a **Route Path** over a barricade like this you will need to add [OffMeshLink](#) to your cover. You can always use the prefabs included as a reference or starting point and just replace the mesh.

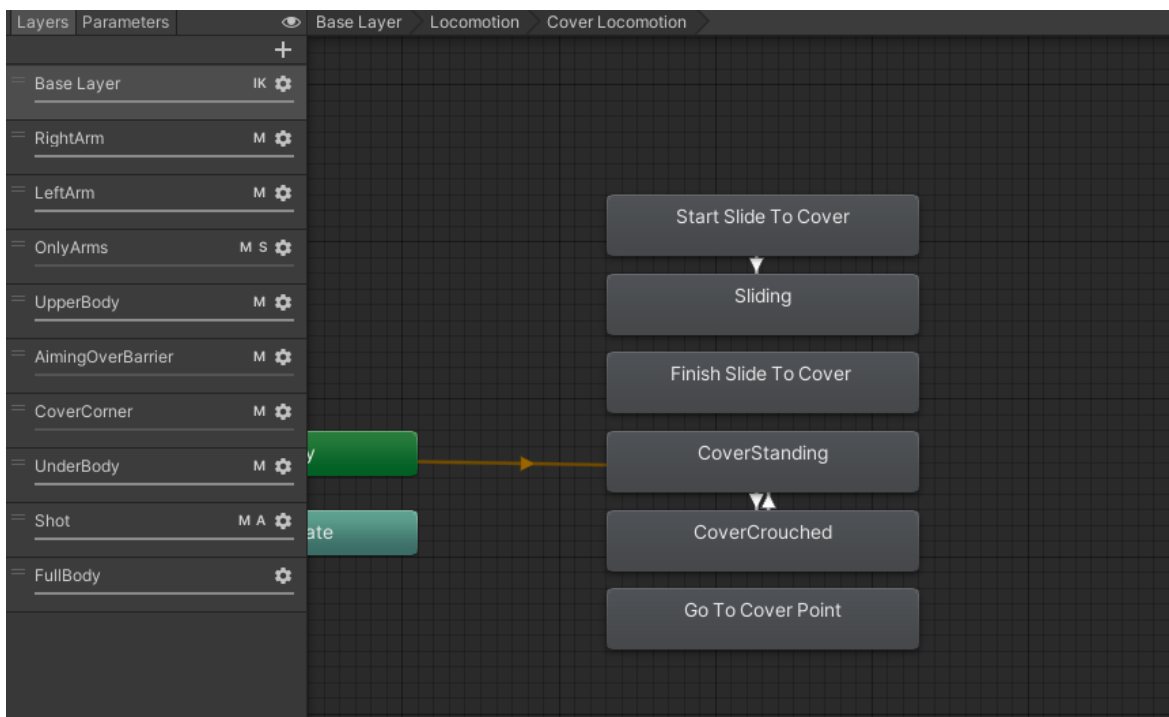


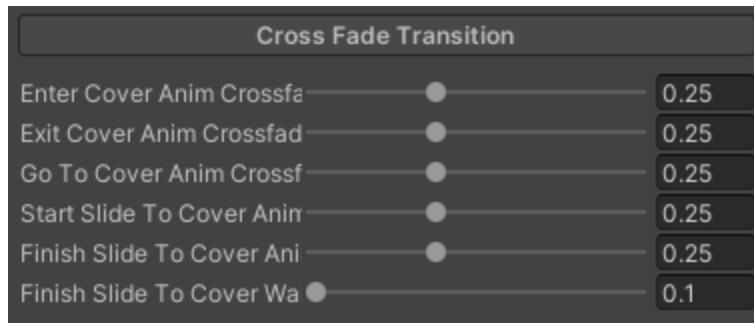
Make sure to add a **JumpDistance** value when you bake the navmesh

- COVER ANIMATION STATES



You don't need to change any of the Default or Cover Locomotion unless you modify the name of the Animation States inside the Cover Animator Controller





To play the Cover animations we use:

animator.[CrossFadeInFixedTime](#)(startSlideToCoverPoint, startSlideToCoverAnimCrossfade);

So each animation has its own crossfade or smooth values and you can set up different values for each animation.

- COVER CAMERA STATES

Cover Settings	Movement Speed	Layers and Tags
Cover UI	Animation States	Camera Settings
Events	Debug	

Camera Offset Right

Corner Left_Offset Camera Right	1
Corner Right_Offset Camera Right	1
Corner Left Aiming_Offset Camera Right	1
Corner Right Aiming_Offset Camera Right	1

Moving to Next Cover CameraState

Go To Cover Camera State	GoToCover
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Stand Cover CameraState

Stand Camera State	CoverStand
Stand Aiming Camera State	CoverStandAiming

Crouch Cover CameraState

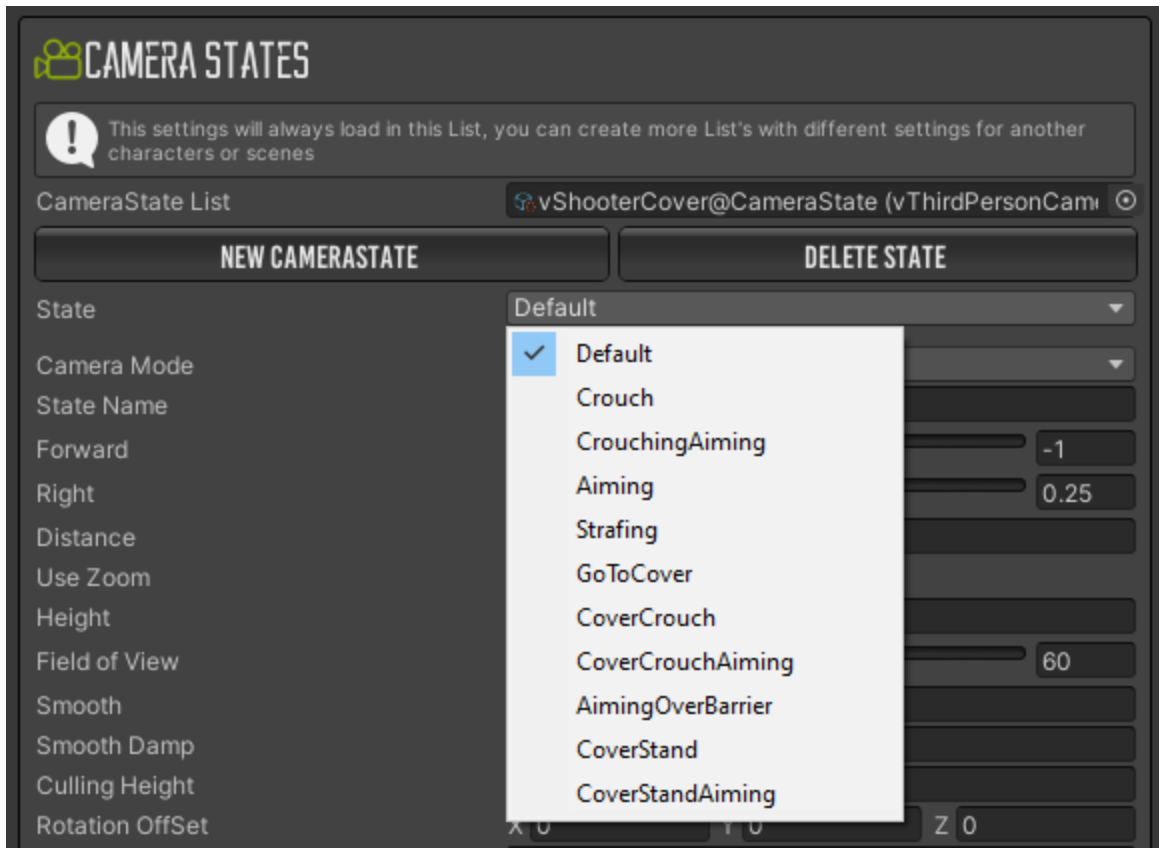
Crouch Camera State	CoverCrouch
Crouch Aiming Camera State	CoverCrouchAiming

HipFire Cover CameraState

Aiming Over Barrier Camera State	AimingOverBarrier
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The Corner Left/Right Camera Offset Right will apply a Offset to the CameraRight value of the current CameraState, increase or decrease this value to have an offset when on a Corner, this goes for Aiming or not Aiming.

The rest of the string values are the actual CameraStates that you can find on your vThirdPersonCamera.



You can always play with the camera states and values to find that sweet spot for a perfect camera cover view.

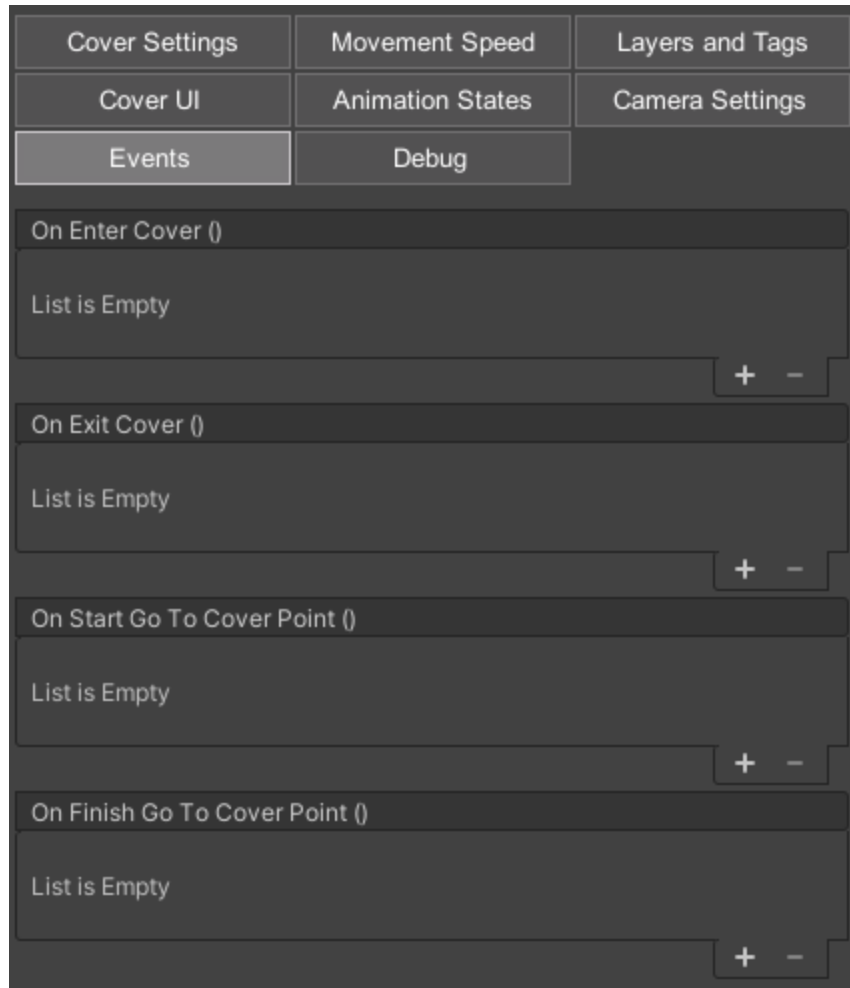
- COVER IK CUSTOM STATES

If the default animations doesn't fit well into your character model, you can always replace the animation or create custom IK adjustments with our IK Adjust feature.



Go to PlayMode and go to the specific condition ex: *Crouching Corner Aiming* and open the ShooterManager > IK Adjust > Edit IK AdjustList, now you can edit the IK of that pose and so on for each Custom IK Pose.

- COVER EVENTS



Here are some basic public Events that you can use to trigger sounds or call public methods for example.

- COVER DEBUG MODE



The Debug is a very powerful tool to use, it will inform you quite a lot of useful information like if it's in cover, corner, what side, what coverpoint, if it's aiming, crouching or standing, etc...

The **Raycasts** for the **Crouch Detection** will also show up when the Debug Mode is enabled.